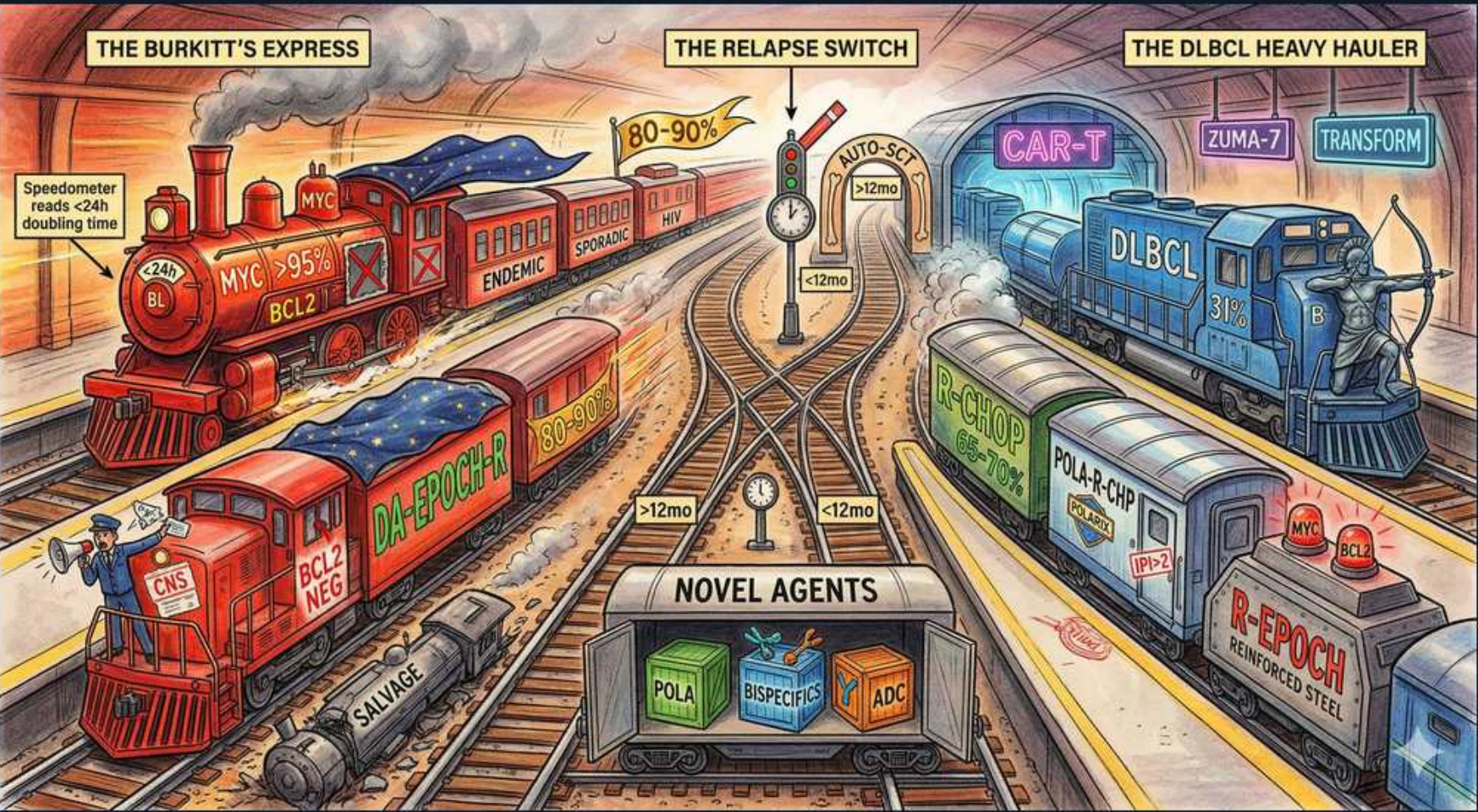


NHL — Burkitt's Express (railway station)





1. MYC engine

WHAT YOU SEE

Red Burkitt engine with 'MYC ~95%'

WHAT IT ENCODES

Burkitt lymphoma is driven by translocation of the c-MYC oncogene on chromosome 8 to an immunoglobulin enhancer, most often t(8;14)(q24;q32) placing MYC under the IgH heavy-chain promoter (~80% of cases); variants are t(2;8) (kappa) and t(8;22) (lambda light chains). MYC is a master transcription factor that drives the cell cycle, so constitutive overexpression produces relentless proliferation. A MYC rearrangement is present in roughly 95% of Burkitt cases and is the central diagnostic feature. Mnemonic: Burkitt is the MYC train — '8;14, MYC takes flight.'

BOARD TRAP

A board stem describing a translocation but naming t(14;18)/BCL2 is follicular lymphoma, and t(11;14)/cyclin D1 is mantle cell — neither is Burkitt. Burkitt is MYC-driven (8;14), and an isolated MYC rearrangement with high Ki-67 in a young patient with an abdominal or jaw mass should point you to Burkitt, not DLBCL.

2. Speedometer / doubling time

WHAT YOU SEE

'Speedometer reads <24h doubling time'

WHAT IT ENCODES

Burkitt is the fastest-growing human tumor, with a doubling time often under 24 hours and a Ki-67 (MIB-1) proliferation index approaching 100% — essentially every cell is cycling. This explosive turnover is why patients can present acutely over days to weeks and why bulky disease carries an extreme risk of spontaneous and treatment-induced tumor lysis syndrome. The near-100% Ki-67 is a classic distinguishing feature versus the lower indices of indolent lymphomas (follicular ~<20%). Mnemonic: Burkitt 'burns' through cell cycles — Ki-67 ~100%.

BOARD TRAP

Choosing a low/intermediate Ki-67 for a starry-sky, MYC+ lymphoma is the trap; Burkitt's Ki-67 is essentially 100%. A Ki-67 well below 100% in an aggressive B-cell lymphoma with MYC rearrangement should make you consider high-grade B-cell lymphoma (double-hit, MYC + BCL2/BCL6) rather than classic Burkitt.

3. BCL2-negative tag

WHAT YOU SEE

The word BCL2 stenciled on the side of the red Burkitt engine but struck/crossed — the engine runs on MYC alone, so the BCL2 tag is there only to be canceled: classic Burkitt is BCL2-negative.

WHAT IT ENCODES

Classic Burkitt is BCL2-NEGATIVE (or only weakly positive): it is a pure MYC-driven tumor without BCL2 overexpression, which contrasts with BCL2+ follicular lymphoma. This matters because a lymphoma that carries BOTH a MYC and a BCL2 (and/or BCL6) rearrangement is a 'double-hit / high-grade B-cell lymphoma,' a distinct, aggressive entity that behaves worse than DLBCL and is NOT classic Burkitt. The starry-sky pattern (sheets of monomorphic blue blasts studded with pale tingible-body macrophages clearing apoptotic debris) plus BCL2-negativity and ~100% Ki-67 nails the Burkitt diagnosis.

BOARD TRAP

WRONG to call a MYC+/BCL2+ lymphoma 'Burkitt.' The discriminator is BCL2: classic Burkitt is BCL2-negative. A MYC+ AND BCL2+ tumor is double-hit high-grade B-cell lymphoma — a separate, poor-prognosis entity requiring DA-EPOCH-R, not the Burkitt label.

4. Endemic / sporadic / HIV

WHAT YOU SEE

'ENDEMIC' 'SPORADIC' 'HIV' boxcars

WHAT IT ENCODES

Three epidemiologic forms of Burkitt. Endemic (equatorial Africa, malaria-belt): EBV-positive in ~95%, peaks in children, classically a JAW or facial bone mass. Sporadic (Western, the form tested most in US boards): EBV-positive in only ~20%, presents as an abdominal/ileocecal mass (right lower quadrant), can mimic appendicitis or intussusception. Immunodeficiency-associated: HIV/AIDS (often at relatively preserved CD4 counts) and other immunosuppressed states. Mnemonic: Endemic = jaw + EBV + Africa; Sporadic = abdomen.

BOARD TRAP

A child in Africa with a JAW mass is endemic (EBV-driven) Burkitt — do not default to 'sporadic.' Conversely, a US patient with a rapidly enlarging abdominal/ileocecal mass is sporadic Burkitt, where EBV is usually NEGATIVE, so a stem hinging on EBV positivity to confirm sporadic Burkitt is the trap.

5. DA-EPOCH-R train

WHAT YOU SEE

Red lower 'DA-EPOCH-R' train, BCL2-NEG

WHAT IT ENCODES

Burkitt requires INTENSIVE, dose-dense multi-agent chemoimmunotherapy because of its explosive kinetics: standard regimens are DA-EPOCH-R (dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, doxorubicin, rituximab) or CODOX-M/IVAC (the Magrath regimen) plus rituximab. CNS prophylaxis with intrathecal methotrexate/cytarabine is mandatory given high CNS relapse risk. Therapy must be paired with tumor lysis prophylaxis (hydration + rasburicase). Mnemonic: Burkitt rides the fast train — EPOCH or CODOX, never the slow R-CHOP local.

BOARD TRAP

Standard R-CHOP is INADEQUATE for Burkitt and is the classic wrong answer — its dosing and CNS coverage are insufficient for a tumor doubling in under 24 hours. Forgetting intrathecal CNS prophylaxis is the second trap; Burkitt has high CNS tropism and untreated CNS disease drives relapse.

6. TLS / rasburicase

WHAT YOU SEE

The SALVAGE supply car on the lower track loaded with hydration/urate-lowering gear — the tumor-lysis prophylaxis wagon that must couple to the fast Burkitt train.

WHAT IT ENCODES

Tumor lysis syndrome (TLS) is the major acute emergency: massive cell turnover releases potassium, phosphate, and nucleic acids, producing HYPERkalemia, HYPERphosphatemia, HYPERuricemia, and secondary HYPOcalcemia, risking acute kidney injury and arrhythmia. Prophylaxis is aggressive IV hydration plus a urate-lowering agent: rasburicase (recombinant urate oxidase, e.g., 0.2 mg/kg IV) for high tumor burden/high uric acid, or allopurinol for lower risk. Avoid urinary alkalization with rasburicase. Monitor electrolytes, uric acid, and creatinine closely. Mnemonic: high-high-high-low — K, PO4, urate up; Ca down.

BOARD TRAP

Choosing allopurinol for a high-tumor-burden Burkitt with already-elevated uric acid is the trap — allopurinol only blocks new urate formation and cannot clear the existing load, so rasburicase is preferred when uric acid is high or TLS is established. A second trap: rasburicase is contraindicated in G6PD deficiency (causes hemolysis/methemoglobinemia).



## 7. 80-90% cure banner

### WHAT YOU SEE

'80-90%' flag

### WHAT IT ENCODES

The curability paradox: despite being the most aggressive, fastest-doubling human malignancy, Burkitt is HIGHLY CURABLE with intensive chemoimmunotherapy — cure rates ~80-90% in children and good-risk adults (lower with advanced age, CNS or bone-marrow involvement, or poor performance status). The same high proliferative fraction that makes it dangerous also makes it exquisitely chemosensitive. Mnemonic: fast to grow, fast to go — aggressive but curable.

### BOARD TRAP

Assuming 'aggressive = incurable' is the conceptual trap. Aggressive B-cell lymphomas (Burkitt, DLBCL) are CURABLE with chemo, whereas indolent lymphomas (follicular, mantle cell) are typically incurable but slow — the prognosis logic is inverted from solid tumors.

## 8. Relapse switch >12mo

### WHAT YOU SEE

'AUTO-SET' switch '>12mo' track

### WHAT IT ENCODES

For aggressive B-cell lymphoma (DLBCL), the timing of relapse after first-line therapy steers second-line choice. LATE relapse (>12 months from completing R-CHOP) generally remains chemosensitive, so the path is salvage chemotherapy (e.g., R-ICE, R-DHAP) followed by high-dose chemo with AUTOLOGOUS stem-cell transplant in responders. Mnemonic: late train (>12mo) → salvage chemo + auto-SCT.

### BOARD TRAP

Reflexively sending every relapsed DLBCL to CAR-T is the trap. For late (>12mo) chemosensitive relapse, salvage chemo + autologous transplant is still standard and curative — CAR-T is reserved for early/primary-refractory disease.

## 9. Relapse switch <12mo

### WHAT YOU SEE

'<12mo' track

### WHAT IT ENCODES

EARLY DLBCL relapse (within 12 months) or primary-refractory disease behaves as chemo-resistant. Landmark trials ZUMA-7 (axicabtagene ciloleucel) and TRANSFORM (lisocabtagene maraleucel) showed anti-CD19 CAR-T improves event-free survival over salvage chemo + autologous transplant in this group, making CAR-T the preferred second-line option. Mnemonic: early train (<12mo) → straight to CAR-T.

### BOARD TRAP

Routing an early-relapsing/primary-refractory DLBCL to autologous transplant is the trap — these tumors are chemo-resistant and do poorly with the old salvage-then-auto pathway. Per ZUMA-7/TRANSFORM, CD19 CAR-T is the better second-line choice for relapse under 12 months.

## 10. DLBCL Heavy Hauler

### WHAT YOU SEE

Blue 'DLBCL' locomotive

### WHAT IT ENCODES

Diffuse large B-cell lymphoma is the MOST COMMON aggressive NHL (~30% of all NHL) and the most common NHL overall. It presents as a rapidly enlarging nodal or extranodal mass, often with B symptoms and elevated LDH. First-line R-CHOP cures roughly 60% of patients; cell-of-origin (germinal-center B-cell vs activated B-cell) and double-hit status refine prognosis. Mnemonic: DLBCL = the big blue hauler, R-CHOP is its standard engine.

### BOARD TRAP

Calling follicular lymphoma the 'most common aggressive' NHL is the trap — FL is the most common INDOLENT NHL; DLBCL is the most common aggressive one. Also, a transformed follicular lymphoma can present as DLBCL, so a rapidly growing node with rising LDH in a known FL patient should prompt re-biopsy.

## 11. CAR-T / ZUMA-7

### WHAT YOU SEE

'CAR-T' 'ZUMA-7' 'TRANSFORM' signs

### WHAT IT ENCODES

Anti-CD19 chimeric antigen receptor T-cell therapy (axicabtagene ciloleucel, lisocabtagene maraleucel, tisagenlecleucel) reprograms the patient's T cells to target CD19. Based on ZUMA-7 and TRANSFORM it is now second-line for early-relapsed/refractory DLBCL. Key toxicities to recognize: cytokine release syndrome (fever, hypotension — treat with tocilizumab/IL-6 blockade ± steroids) and ICANS (immune effector cell-associated neurotoxicity — managed with steroids). Mnemonic: CAR-T fires, CRS and ICANS are the recoil.

### BOARD TRAP

Mistaking CAR-T-related CRS for sepsis and withholding tocilizumab is a management trap. High fever and hypotension shortly after CAR-T infusion is CRS until proven otherwise; tocilizumab (anti-IL-6R) is the targeted treatment, alongside supportive care.

## 12. R-CHOP car

### WHAT YOU SEE

Green 'R-CHOP' boxcar

### WHAT IT ENCODES

R-CHOP is the backbone for DLBCL: Rituximab (anti-CD20), Cyclophosphamide, Hydroxydaunorubicin/doxorubicin (the 'H'), Vincristine (Oncovin, the 'O'), and Prednisone, given every 21 days for ~6 cycles. Doxorubicin is cardiotoxic (check ejection fraction; cumulative dose limit), vincristine causes peripheral neuropathy, and rituximab mandates hepatitis B screening before use. Mnemonic: CHOP = Cyclophosphamide, Hydroxydaunorubicin, Oncovin, Prednisone; add R for Rituximab.

### BOARD TRAP

Giving R-CHOP without screening for hepatitis B is the safety trap — rituximab can reactivate HBV and cause fulminant hepatitis. Also do not use R-CHOP for Burkitt; its intensity and CNS coverage are inadequate for that disease.



### 13. Pola-R-CHP

#### WHAT YOU SEE

'POLA-R-CHP' boxcar

#### WHAT IT ENCODES

Pola-R-CHP (from the POLARIX trial) substitutes the antibody-drug conjugate polatuzumab vedotin (anti-CD79b–MMAE) for vincristine, giving Polatuzumab + Rituximab + Cyclophosphamide + Doxorubicin (H) + Prednisone. It improved progression-free survival versus R-CHOP in higher-risk DLBCL (IPI 2-5), making it a frontline option for those patients. Mnemonic: drop the O (vincristine), add Pola — 'Pola swaps the Oncovin.'

#### BOARD TRAP

Assuming Pola-R-CHP replaces R-CHOP for everyone is the trap — its benefit was shown mainly in higher-risk (IPI  $\geq 2$ ) DLBCL, and R-CHOP remains standard for many. Polatuzumab adds its own neuropathy risk (it is a microtubule-inhibitor conjugate).

### 14. Novel agents (POLA/ADC)

#### WHAT YOU SEE

'NOVEL AGENTS — POLA, BISPECIFICS, ADC'

#### WHAT IT ENCODES

The relapsed/refractory aggressive-NHL toolkit beyond chemo: antibody-drug conjugates (polatuzumab vedotin anti-CD79b, loncastuximab tesirine anti-CD19), CD20xCD3 bispecific antibodies (glofitamab, epcoritamab) that bridge T cells to tumor, CD19 CAR-T, and tafasitamab (anti-CD19) + lenalidomide. Bispecifics, like CAR-T, can cause cytokine release syndrome. Mnemonic: when chemo fails, redirect the immune system — ADCs deliver the toxin, bispecifics/CAR-T deliver the T cell.

#### BOARD TRAP

Forgetting that bispecific antibodies also cause CRS (not just CAR-T) is the trap — glofitamab/epcoritamab require step-up dosing and CRS monitoring. Treating CRS from a bispecific as infection alone, rather than with tocilizumab/steroids when indicated, is the management error.